MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FOURTH SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

MG4DSEBCA200- Data Visualization

Duration of the Examination: **1.5 Hrs** Maximum Marks: **35**

List of Questions

***** All questions belong to the **Apply** level and are aligned with Course Outcome 3 (**CO3**).

1. Prepare a basic **sales performance report** for a small retail store using **Microsoft Power BI**. Follow the steps below to complete the task:

   a) Install **Power BI Desktop** (if not already installed) and open a new Power BI project file.

   b) Identify and describe the functions of the **Ribbon**, **Panes (Fields, Visualizations, Filters)**, and **Canvas** in the Power BI workspace.

   c) Import the dataset **SalesData.xlsx** and **Products.csv**

   d) Establish a relationship between the two datasets using **ProductID** as the common key.

   e) Create a simple report with the following visuals:

      i. A **bar chart** showing total sales by product category.

      ii. A **card visual** showing total sales amount.

      iii. A **table visual** displaying product name, quantity sold, and total amount.

   f) Save the report as **Retail_Sales_Report.pbix**.

2. Import the dataset **EmployeeData.xlsx** and identify at **least 3 issues** using **Power Query Editor Interface.**

3. Import the dataset **CustomerDetails.csv** into Power BI. In **Power Query Editor**, perform the following **cleaning operations**:

   a) Remove **duplicate rows**.

   b) Replace missing or null values in the **PhoneNumber** column with "Not Provided".

   c) Remove any unwanted blank rows.

4. Import the dataset **Sales_Regions.csv** (with a “FullName” and “SalesValue” column). Use **Power Query** to:

a) **Split** the FullName column into FirstName and LastName.

b) **Change** the data type of SalesValue from Text to Decimal Number.

c) **Rename** the column SalesValue to TotalSales.

5. To combine data from multiple sources, import two datasets: **Sales_Q1.xlsx** and **Sales_Q2.xlsx** and do the following:

a) Use **Append Queries** to combine both tables into a single dataset named Sales_2024.

b) Import Product_Details.xlsx and **Merge** it with Sales_2024 using the common column ProductID.

c) Load the final combined dataset to Power BI.

6. To derive new insights using **custom and computed fields**, do the following:

Using the merged dataset (**Sales_2024**), create:
A **Custom Column** in Power Query named Profit using the formula:
A **Calculated Column** (in Data View) named DiscountedPrice using DAX:
Add a Card visual in the report to show **Total Profit**.

7. Import the datasets given below into Power BI

`Sales_2024.xlsx` (columns: SaleID, ProductID, CustomerID, Quantity, TotalSales)

`Product_Details.xlsx` (columns: ProductID, ProductName, Category, Cost)

`CustomerDetails.csv` (columns: CustomerID, CustomerName, City, Region)

Perform the following:
Create relationships between them using **ProductID** and **CustomerID** as key fields.
View and edit the relationships in the **Model view** (one-to-many or many-to-one).
Create a **pie chart** showing total sales of each product.
Create a **bar chart** showing total sales by region.

8. Import the datasets given below into Power BI

Sales_2024.xlsx
Product_Details.xlsx

Perform the following:
Check if all ProductIDs in Sales_2024 exist in Product_Details.
Identify any missing or mismatched records.
Use **Power Query Editor** to remove or correct inconsistent ProductIDs.

9. Use the **Sales_2024** table which contains Quantity, Cost, and TotalSales fields to do the following:

a) Create a calculated column named Profit using the formula

Profit = [TotalSales] - ([Quantity] * [Cost])
Add another calculated column **DiscountedSales = [TotalSales] * 0.9**
Display both Profit and DiscountedSales in a **Table visual**.

10. Create the following **measures** using the **Sales_2024** table:

a) Total Profit = SUM(Sales_2024[Profit])
b) Average Profit per Sale = AVERAGE(Sales_2024[Profit])
c) Display both measures using a Multi-row Card visual.

11. Remove **inconsistencies** in the data:
    a) Replace the **missing values** [CustomerName = (blank)]
    b) Remove two **identical** sales records.
    c) **Change data type** of date stored as text.
    d) **Capitalize** each region.
    e) **Trim** extra spaces in name field.
    f) Fix missing keys.
    g) **Filter** the outliers [e.g., Quantity = -5]
    h) **Change** date **formats**.
12. Create a **bar chart** showing *Total Sales by Product*.
    • Add a descriptive **title**.
    • Change **title font**, **size**, and **color**.
    • Apply a **background color** or **border**.
    • Format the bar colors based on sales value (gradient colors).
13. Create a **clustered column chart** showing *Sales by Category*.
    Change the column color and font style of labels.
    Sort visuals by *Sales* (descending).
    Apply a **filter** to show only *Electronics* and *Furniture* categories.
    Apply **conditional color formatting** to Sales column.
    • Sales > 50,000 → Green
    • Sales ≤ 50,000 → Red
14. Create a **line chart** showing *Sales Trend over Time*.
    Format the line color to blue and enable data markers.
    Sort the chart by Month.
15. Create a **pie chart** to display *Profit share by Category*.
    Add data labels to show percentages.
    Reposition the legend.
    Change color of each slice.
16. Create **an area chart** showing Quantity sold over time.
    Apply transparency and change area color to green.
17. Create a **combo chart** *"Sales vs Profit by Category"* with: [**Column chart:** Total Sales, **Pie chart:** Profit, **Line chart:** Quantity, Axis → Category]
    Customize the chart: Add data labels and title "Sales vs Profit by Category".
    Use visual interactions (clicking one visual filters another